

TA Class 4

Money Demand, the Price Level and New Keynesian AS–AD

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Today

- 1 Money market – From identity to theory and Baumol-Tobin example
- 2 Monetary policy, prices and inflation
- 3 **Benigno (2015)**: New Keynesian Economics AS-AD view

Theories of Money

An Identity Is Not a Theory

$$MV \equiv PY \quad (1)$$

- M , P and Y are measured independently, while V **is not**: velocity is *defined* as $V \equiv PY/M$
- What we have is an **ex-post fact** rather than an **ex-ante mechanism** explaining the co-movement of these variables
- We require assumptions or other structure to make this into a theory
- **Note**: Money M is **neither demand or supply**, but rather the amount held in equilibrium

What Would Turn It Into a Theory

- 1 **A partition:** Say which letters are taken as given and which adjust. The **Quantity Theory** takes Y as determined by real forces and V as stable, and lets P adjust to M
- 2 **A behavioural claim.** “ V is stable” is an **assumption about agents behavior**. It does not follow from the algebra
- 3 **A refutable comparative static.** With 1 and 2, $\Delta \ln P = \Delta \ln M$ for given Y : a prediction the data can contradict

Cambridge Equation

$$\frac{M^D}{P} = k Y, \quad k \equiv 1/V \quad (2)$$

k is now a *desired* ratio, something an agent chooses, and therefore something a model can be asked to deliver

- But what if k (or V) is not constant? What determines it?

Money Market Equilibrium: Which Side Carries the Content?

$$M^S = m^D(Y, i) P \quad (\text{K 11.2.1})$$

- Usually assume **supply** is influenced by the central bank.
- Demand is chosen by household based on their choice variables **demand** that households choose. The identity cannot tell them apart
- If M^S rises, exactly three margins can restore equilibrium: $p \uparrow$, $i \downarrow$, $Y \uparrow$. **The equation does not choose between them** – Theory decides what moves and how
- **The Quantity Theory's answer:** All power is in the supply: Y is real-determined and m^D is rigid, so P absorbs changes in M

Money Market Equilibrium: Which Side Carries the Content?

- **The alternative:** Model the demand side and let it tell us how much of the adjustment runs through (Y, i) rather than through P
- **(Keynes (1936, ch. 15): Why do people hold money?)**
 - 1 **Transactions:** Money is held to pay for things. Income arrives in lumps and spending is spread out, so a balance has to be carried in between
 - 2 **Precautionary:** Money is held against the unexpected – a payment that arrives early, a bill that is larger than planned, an opportunity that has to be taken now
 - 3 **Speculative:** Money is held as an asset rather than as a means of payment: when other assets look likely to lose value, staying in cash is the safer position
- Baumol-Tobin focuses on the first motive and formalizes a theory around it.

Baumol-Tobin: The Ingredients

Baumol (1952) and Tobin (1956): a household problem whose *solution* is a money demand function

- 1 **Two assets.:** Money pays no interest; everything else pays the nominal rate i . (holding money is costly)
- 2 **Payment technology:** All purchases must be settled in money
- 3 **Timing:** The household spends pY evenly over the period, with perfect foresight.
(This is what makes the average balance computable)
- 4 **Friction:** Converting interest-bearing assets into money costs a fixed real amount F per trip – Think of it as fees, time, attention.
- 5 **Choice variable:** The number of trips $N \rightarrow$ Will determine money demand

Read the assumptions as a design, not as realism

Drop the first, the third or the fourth and there is no interior N – and therefore no money demand function. That is the checklist to run on every model you meet: which assumption is load-bearing, and which is decoration?

Baumol-Tobin: The Trade-off

With N evenly spaced trips, balances saw-tooth between $P\frac{Y}{N}$ and 0, so the average nominal balance is

$$M = \frac{pY}{2N} \quad (\text{K 10.4.1})$$

The household minimises the total cost of transacting:

$$\min_N \underbrace{pFN}_{\text{cost of the trips}} + \underbrace{i \frac{pY}{2N}}_{\text{forgone interest}} \quad (\text{K 10.4.2})$$

First-order condition, and the solution:

$$pF - \frac{i p Y}{2} N^{-2} = 0 \quad \Rightarrow \quad N^* = \sqrt{\frac{i Y}{2 F}}, \quad \Rightarrow \quad \frac{M}{p} = \sqrt{\frac{Y F}{2 i}} \quad (\text{K 10.4.3-4})$$

What the Model Buys, and Where It Stops

- **It earns a downward-sloping money demand schedule** which is derived rather than assumed. Giving us an interpretation for its shifts: anything that moves F moves the whole schedule
- **It defines a cost of inflation for households** (crucial for welfare analysis) $\pi \uparrow \Rightarrow i \uparrow \Rightarrow N \uparrow$, and the total cost of trips is $NF = \sqrt{(r + \pi)YF/2}$
- **It covers only the transactions motive.** Keynes's precautionary and speculative motives are absent
- **F is a black box**, and it is exactly the object technology moves: ATMs, debit cards, Pix. So is the boundary of “money”, which is an accounting convention and not a stable one

Money, Prices and Inflation

What We Are Assuming

$$M^S = m^D(Y, i) \cdot p$$

- **A1. Prices are flexible.** p moves at every date to clear this equation. Nothing else is available to do the job
- **A2. The real side is determined elsewhere.** Y and r come from the real blocks we built earlier and do not depend on M^S or p . (classical dichotomy)
- **A3. Money demand is a stable schedule** $m^D(Y, i)$, increasing in Y and decreasing in i , similar to Baumol-Tobin, but not microfounded
- **A4. Fisher, with perfect foresight.** $i = r + \pi$: the *nominal* rate carries inflation, the *real* rate does not move

What is left to determine

This is a steady-state view of money market equilibrium.

With Y and r fixed, this is one equation in two nominal objects, p and π : anything money does must show up in one of them.

Steady State with a Constant Money Supply

Y and r constant, and the central bank holds M^S fixed

- **Conjecture:** the price level is constant as well
- Then inflation is zero, so $i = r$, and the money market clears at

$$p = \frac{M^S}{m^D(Y, r)}$$

- which is indeed constant, because M^S , Y and r all are. **The conjecture is confirmed**
- **A one-time rise in M^S** simply moves us between two such steady states, so $p'/p = M^{S'}/M^S$: prices jump in proportion, once, and nothing real changes

The intuition

People want a certain amount of *real* balances. Hand them more nominal money and they all try to spend it down – but somebody has to hold it. The only way out is for money to be worth less

Steady State with a Growing Money Supply

- **Conjecture:** inflation is constant at μ
- If $\pi_{t+1} = \mu$, then $i_{t+1} = r + \pi_{t+1} = r + \mu$ is constant, and so are desired real balances

$$\begin{aligned}M_t^S &= m^D(Y, r + \mu) p_t \\M_t^S(1 + \mu) &= m^D(Y, r + \mu) p_t(1 + \mu) \\M_{t+1}^S &= m^D(Y, r + \mu) p_{t+1}\end{aligned}$$

- The market clears in $t + 1$ as well.
- **The conjecture is confirmed:** sustained money growth shows up one-for-one in inflation
- **Assuming** that the economy is not growing; otherwise, part of the new money is absorbed by money demand: $\pi = \mu - \eta g$ (K 11.2.3), where η is the income elasticity of money demand

Monetary Policy and the Price Level

Starting Again From the Money Market

$$M^S = m^D(Y, i) p$$

- Real variables are determined elsewhere. So the central bank determines the **price level** by choosing M^S , and determines **inflation** by choosing the growth rate of M^S
- That is a strong and attractive claim: nominal outcomes are a policy choice, and only a policy choice
- It also delivers the long-run results you want a first course to leave you with – money neutrality, $\pi = \mu - \eta g$, seignorage as a tax on real balances, the Friedman rule

The problem

No central bank has operated this way for forty years. None of them targets a monetary aggregate; all of them announce a short-term nominal interest rate. So the mechanism in the box above is not the mechanism actually at work

Why the Instrument Changed

Poole (1970): the choice between M and i is an empirical question, not a doctrinal one

- **If shocks come from spending**, fixing M is a stabiliser: a boom raises money demand, i rises automatically, and the boom is damped. Fixing i lets the boom run
- **If shocks come from money demand**, fixing M is a destabiliser: a shift in F or in the composition of “money” passes one-for-one into i and then into output. Fixing i absorbs it entirely
- **The 1979–82 experiments answered the question.** Money demand moved far more than anyone had assumed – for exactly the reasons the last section gave. Since then the instrument has been the interest rate

Consequence for the rest of the course

Benigno's model, and Galí's, contain no money and no LM curve. That is not a simplification for the blackboard. It is a description of the regime we are in

Setting i : What Gets Lost

Drop M^S from the model and announce an interest rate instead

The equilibrium condition goes with it, and the Fisher equation is what remains:

$$i_t = r + \mathbb{E}_t \pi_{t+1}$$

- A peg, $i_t = \bar{i}$, delivers $\mathbb{E}_t \pi_{t+1} = \bar{i} - r$. **Expected inflation** is pinned down
- **The level of prices is not.** Any p_0 , and any path whose expectation satisfies the equation, is an equilibrium. There is a continuum of them (Sargent and Wallace, 1975)
- Notice the bookkeeping: we removed an equation, $M^S = m^D \cdot p$, and put back a weaker one. The model lost an equation before it lost an instrument, and that is where the indeterminacy came from

Plainly

A central bank that only ever announces a number for the overnight rate has said nothing about where the price level will be

The Taylor Principle

Replace the peg with a rule: commit to *respond*

$$i_t = \rho + \phi_\pi \pi_t \quad \implies \quad \mathbb{E}_t \pi_{t+1} = \phi_\pi \pi_t \quad (\text{at } r = \rho)$$

- $\phi_\pi < 1$: deviations die out on their own, so every π_0 starts a valid bounded path – indeterminacy again
- $\phi_\pi > 1$: any $\pi_0 \neq 0$ explodes and is ruled out, leaving a unique bounded equilibrium
- **The economic content.** Respond *more than one-for-one*, so that when inflation rises the *real* rate rises with it. Anything less and the bank validates whatever inflation the public happens to expect

The shape of the answer

Determinacy is a property of a *rule*, not of an action. What pins down the price level is what the central bank would do in states of the world that never occur – which is also why credibility is part of the mechanism and not a footnote to it

Why Monetary Policy Is Not Simple

- 1 **The instrument is nominal, the transmission is real.** Moving i matters only because something is sticky. Remove the rigidity and we are back to the classical dichotomy, where none of this mattered
- 2 **The relevant object is a path, not a number.** Only the expected sequence of short rates moves the long real rate households respond to
- 3 **The target moves.** The bank must set r relative to an unobserved natural rate r_n that shifts with productivity, spending and beliefs
- 4 **Determinacy requires commitment** – a rule, understood and believed
- 5 **The constraints bind:** zero lower bound, credibility, fiscal backing

Bridge

Galí's book is this list written as a single model. Benigno (2015), which we turn to now, is that model with the dynamics removed so that it fits in a diagram

**Benigno (2015):
New Keynesian Economics from AS-AD View**

The Modelling Framework

Two periods: the short run has sticky prices, the long run does not

- **Households.** $u(C) - v(L) + \beta\{u(\bar{C}) - v(\bar{L})\}$ (1), under a single intertemporal budget constraint linking the two periods (2). Isoelastic u with elasticity $\tilde{\sigma}$; isoelastic v with parameter η , the inverse Frisch elasticity
- **Firms.** Monopolistic competition with demand elasticity θ (9), linear technology $Y(j) = AL(j)$. A fraction α is stuck at a predetermined price P^e ; the remaining $1 - \alpha$ reoptimise
- **Monetary policy** sets i in the short run and anchors $\bar{p}$ in the long run. *There is no money supply anywhere in the model*
- **Fiscal policy** sets $\{g, \bar{g}\}$ and the tax vector $\{\tau_y, \tau_w, \tau_c, \tau_l\}$, with transfers T balancing the government's intertemporal constraint (18)

What the two periods buy

Short run: $\alpha > 0$, money is non-neutral. Long run: $\alpha = 0$, classical dichotomy restored. The whole of monetary non-neutrality lives in one parameter

Five Things to Take Away Before Any Algebra

- 1 **AD is an Euler equation.** It slopes down because a higher *current* price level, with the long-run price level anchored, raises the real interest rate and postpones consumption. No money market, no LM curve
- 2 **AS is a pricing decision.** It slopes up because higher output means more hours, a higher real wage, higher real marginal cost, and therefore higher prices from the firms that can move
- 3 **Price rigidity is the only source of non-neutrality.** Let $\alpha \rightarrow 0$ and AS is vertical at y_n
- 4 **Three output concepts, not one:** actual y , natural y_n , efficient y_e . Whether a shock moves them together is what decides whether policy faces a trade-off
- 5 **Optimal policy is one more line in the same picture**

Aggregate Demand: The Derivation Path

Households maximise (1) subject to (2)

1 Intratemporal condition, each period: $v_l(L)/u_c(C) = W/P$ (7)

2 Euler equation: $u_c(C)/[\beta u_c(\bar{C})] = (1+i)P/\bar{P} = 1+r$ (3)

3 Isoelastic u , take logs: $\bar{c} - c = \tilde{\sigma}(r - \rho)$ (4)

4 Write r in terms of the instrument: $\bar{c} - c = \tilde{\sigma}[i - (\bar{p} - p) - \rho]$ (5)

5 Resource constraints $Y = C + G$ and $\bar{Y} = \bar{C} + \bar{G}$, log-linearised: $y = s_c c + g$ and $\sigma \equiv \tilde{\sigma} s_c$

6 Substitute:

Aggregate demand

(6)

$$y = \bar{y} + (g - \bar{g}) - \sigma [i - (\bar{p} - p) - \rho]$$

With consumption taxes, $-\sigma(\bar{\tau}_c - \tau_c)$ enters the bracket (8)

Reading AD

- **The slope is $-1/\sigma$** in (y, p) space. High σ means a small price movement is enough to move saving a lot, so AD is flat
- **It slopes down because of the real interest rate**, not because of a demand for liquidity. That is the whole difference from IS-LM
- **What shifts it:** short-run monetary policy ($i \downarrow$), short-run fiscal policy ($g \uparrow, \tau_c \downarrow$), and the long-run variables $\bar{p} \uparrow$ and $\bar{y}_n \uparrow$
- **$\bar{p}$ is the anchor.** Long-run monetary policy is the choice of $\bar{p}$; short-run policy is the choice of i . The determinacy problem from the last section is handled here by assumption

Why the future sits on the demand curve

Anything that raises expected future consumption raises current consumption through the Euler equation. Expectations are not an add-on to AD – they are most of it

Aggregate Supply: The Derivation Path

Firms price optimally, given the labour supply the household block delivers

1 Demand for variety j : $Y(j) = (P(j)/P)^{-\theta}(C + G)$ (9); profits with taxes τ_y, τ_w (10)

2 Reoptimising firms set a markup over marginal cost, $P(j) = (1 + \tilde{\mu})W/A$, with

$$\tilde{\mu} = \frac{\theta}{\theta - 1} \frac{1 + \tau_w}{1 - \tau_y} - 1 \quad (11)$$

3 Substitute the household's real wage (7) to express the relative price in terms of *quantities* (12)

4 Define y_n : the output that would obtain if *all* firms could adjust (14)–(15)

5 The price index is a weighted average, $p = \alpha p^e + (1 - \alpha)\tilde{p}$, with $\tilde{p} - p = (\sigma^{-1} + \eta)(y - y_n)$ (16)

6 Eliminate $\tilde{p}$:

Aggregate supply (17)

$$p - p^e = \kappa(y - y_n), \quad \kappa = \frac{(1 - \alpha)(\sigma^{-1} + \eta)}{\alpha}$$

Reading AS

- **The slope is about α only.** κ falls with α : more stuck firms means a flatter AS, so a given demand impulse produces more output and less inflation. $\alpha \rightarrow 0$ gives $\kappa \rightarrow \infty$: vertical, classical
- **The position is y_n .** AS always passes through (p^e, y_n) . Every real shock relocates the curve by relocating y_n , and nothing about α is involved
- **$(\sigma^{-1} + \eta)$ is the slope of real marginal cost in output** – σ^{-1} from the consumption side of the labour supply decision, η from the disutility of work. The same object as in the labour block

A warning about what an equation identifies

This is a New *Classical* Phillips curve, not yet a New Keynesian one: unexpected prices, and no $\mathbb{E}_t \pi_{t+1}$. Phelps (1967) and Lucas (1973) derive the same equation from imperfect information rather than from sticky prices. *An equation does not identify its own microfoundation*

Natural Versus Efficient Output

The comparison that organises every comparative static in the paper

Natural

(15)

Efficient

(19)

$$y_n = \frac{1+\eta}{\sigma^{-1}+\eta}a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}g - \frac{\mu}{\sigma^{-1}+\eta}$$

What a flexible-price economy *would* produce, monopoly distortion included

$$y_e = \frac{1+\eta}{\sigma^{-1}+\eta}a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}g$$

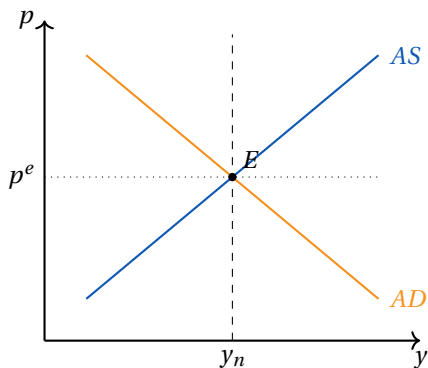
What maximises $u(C) - v(L)$ subject to technology and resources

$$y_n - y_e = -\frac{\mu}{\sigma^{-1}+\eta}$$

And therefore

Productivity and public spending move y_n and y_e by the *same* amount: closing the output gap and stabilising prices become the same objective, so there is *no trade-off*. Mark-up shocks move y_n only: the two objectives now point in different directions, and that is the only reason optimal policy needs a weight at all

The Diagram



- AS passes through (p^e, y_n) by construction (17)
- AD is drawn assuming $\bar{p} = p^e$ and i set at the natural rate, so it crosses y_n too (21)
- So the initial equilibrium E has a zero output gap and stable prices. Everything that follows is a displacement from E

Two questions per shock

Does y_n move? And does y_e move with it? The first tells you where AS goes; the difference tells you whether policy has a choice to make

Four Shocks, One Table

Shock	What moves	Outcome, no policy	Optimal policy
Temporary $a \uparrow$ (Fig. 6)	$y_n \uparrow$, $y_e \uparrow$; AS shifts down	$y \uparrow$ but by less than y_n ; $p \downarrow$; <i>negative</i> gap	$i \downarrow$: AD up, gap closed <i>and</i> prices stable
Permanent $a \uparrow$ (Fig. 7)	y_n and $\bar{y}_n \uparrow$; AS down <i>and</i> AD up	Gap closed and prices stable on its own	Do nothing: r_n is unchanged because both rise in proportion
Expected future $\bar{a} \uparrow$ (Fig. 8)	$\bar{y}_n \uparrow$ only; AD up	y and p both rise; positive gap	$i \uparrow$: lean against the optimism
Temporary $\mu \uparrow$ (Fig. 9)	$y_n \downarrow$, y_e <i>unchanged</i> ; AS up	Stagflation: $y \downarrow$, $p \uparrow$	<i>Trade-off</i> . Price stability wants $i \uparrow$; the efficient level wants $i \downarrow$

The lesson

Monetary policy can stabilise prices and the output gap *simultaneously* – except for mark-up shocks. And the *direction* of the required move depends entirely on whether the shock is temporary, permanent, or merely expected

Optimal Policy as One More Line

A second-order approximation to household utility (Woodford, 2003, ch. 3) gives the objective

$$L = \frac{1}{2}(y - y_e)^2 + \frac{1}{2} \frac{\theta}{\kappa} (p - p^e)^2 \quad (33)$$

Substituting AS (17) and minimising over y gives the targeting rule

$$(y - y_e) + \theta(p - p^e) = 0 \quad (34)$$

- A line through (p^e, y_e) with slope $-1/\theta$: the *IT* schedule, drawn in the same diagram as AS and AD
- **Robust.** The same rule is optimal for every shock, temporary or permanent, whatever its source (Svensson, 2007). Optimal policy is then: move i until the economy sits where IT crosses AS
- A bank that cares more about prices has a *flatter* IT. For productivity and spending shocks IT already passes through the new equilibrium and the trade-off is empty; for mark-up shocks it does not, and the slope of IT is where the policymaker's preferences finally enter

What Carries Into the Next Course

- **AS becomes the New Keynesian Phillips curve**

$$\pi_t = \kappa \tilde{y}_t + \beta \mathbb{E}_t \pi_{t+1}$$

- **AD becomes the dynamic IS equation,**

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \sigma (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n)$$

- **IT becomes the optimal targeting rule**, with the Taylor rule as its operational counterpart – and the Taylor principle from the last section as the condition that makes it work
- **What the two-period version leaves out:** forward-looking price setting and therefore inflation persistence and the cost of disinflation; an explicit process for r_t^n ; commitment versus discretion; financial frictions, heterogeneity, the open economy

Thank you!

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